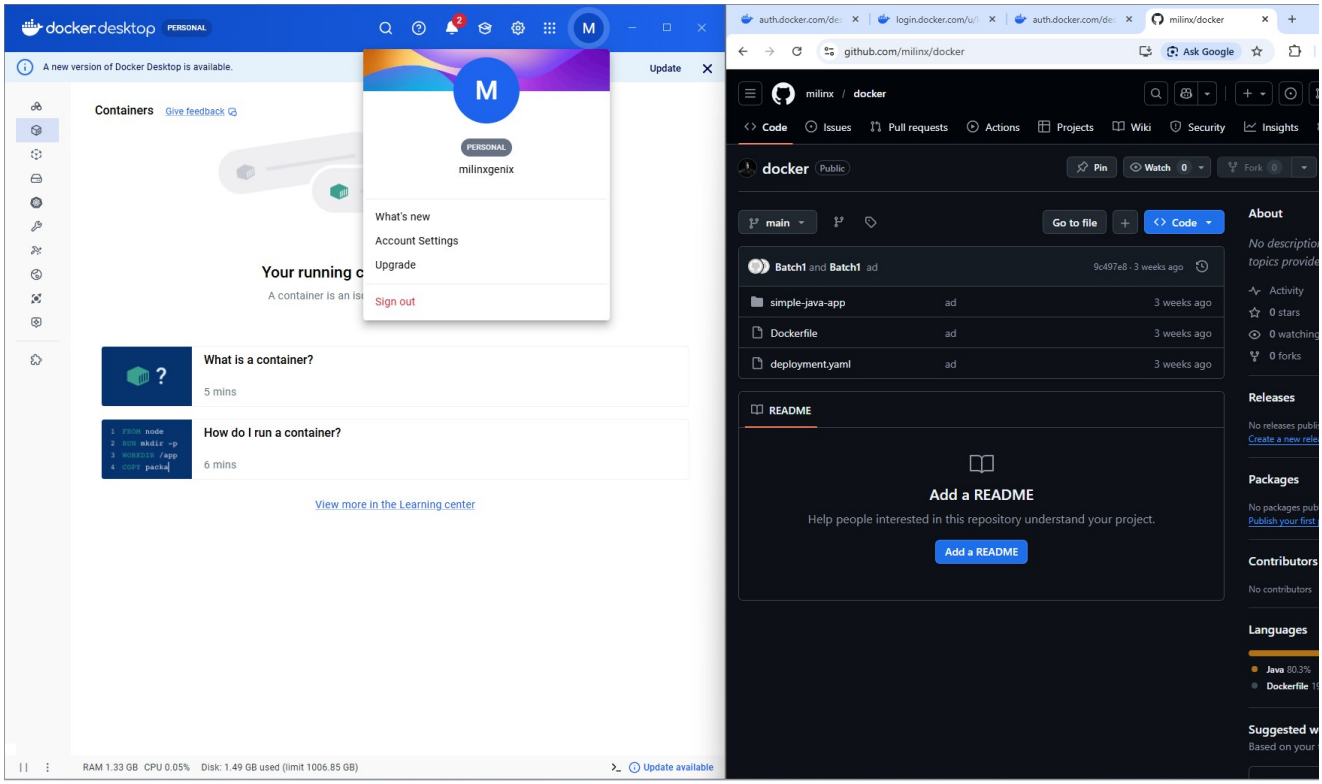
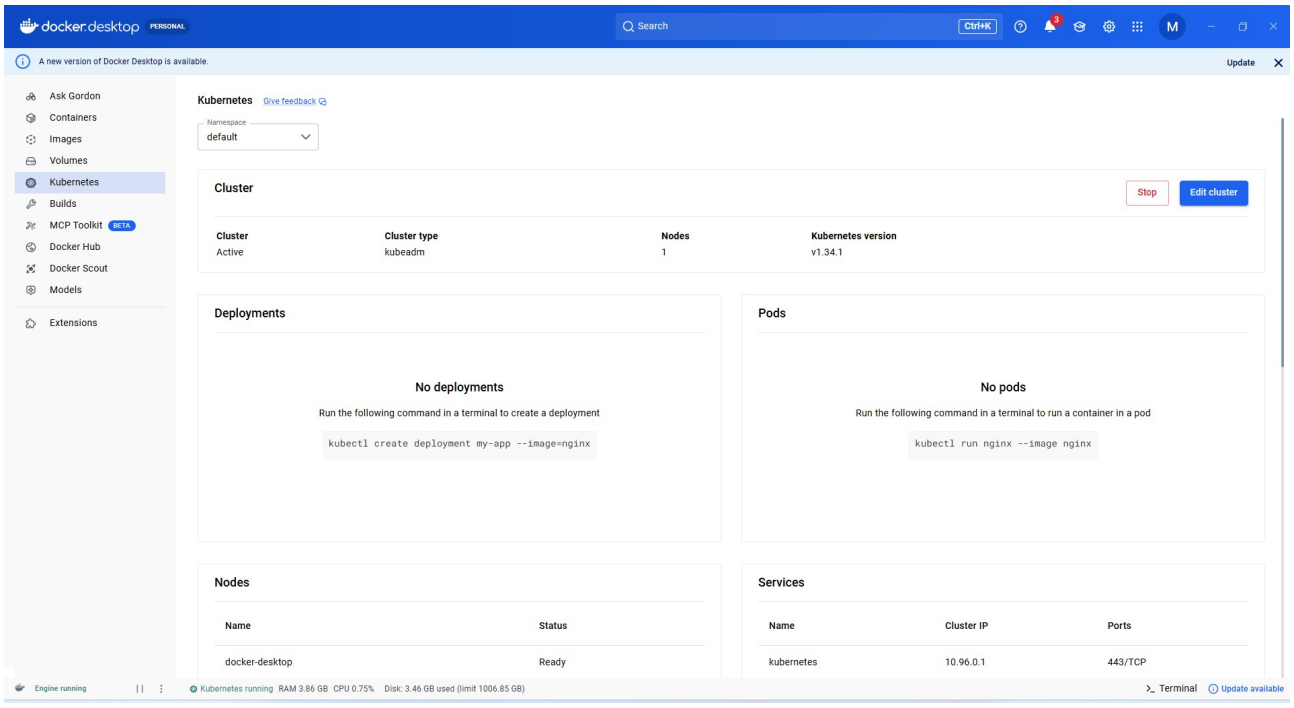
NAME : KUMAR MILIND
REG : 23MIS0650
AGILE LAB ASSESSMENT – 4
(ISWE406P)
DATE : 30th MARCH

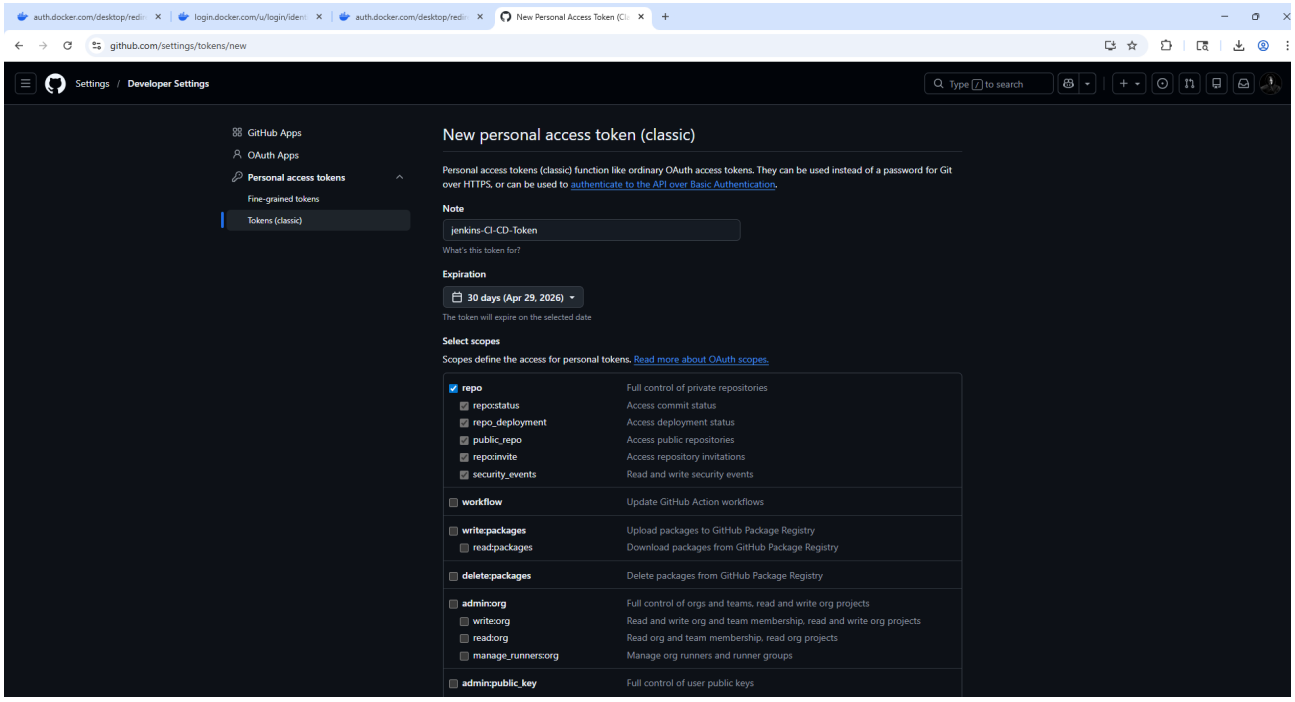
DOCKER & GITHUB SETUP



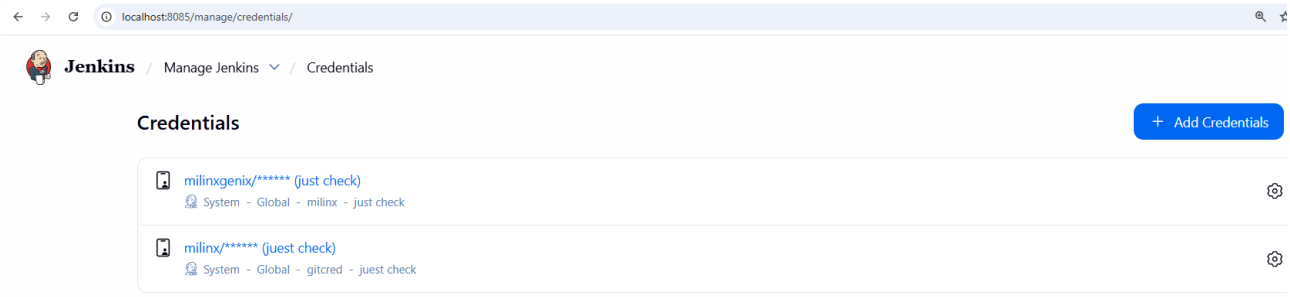
KUBERNETES ENABLED



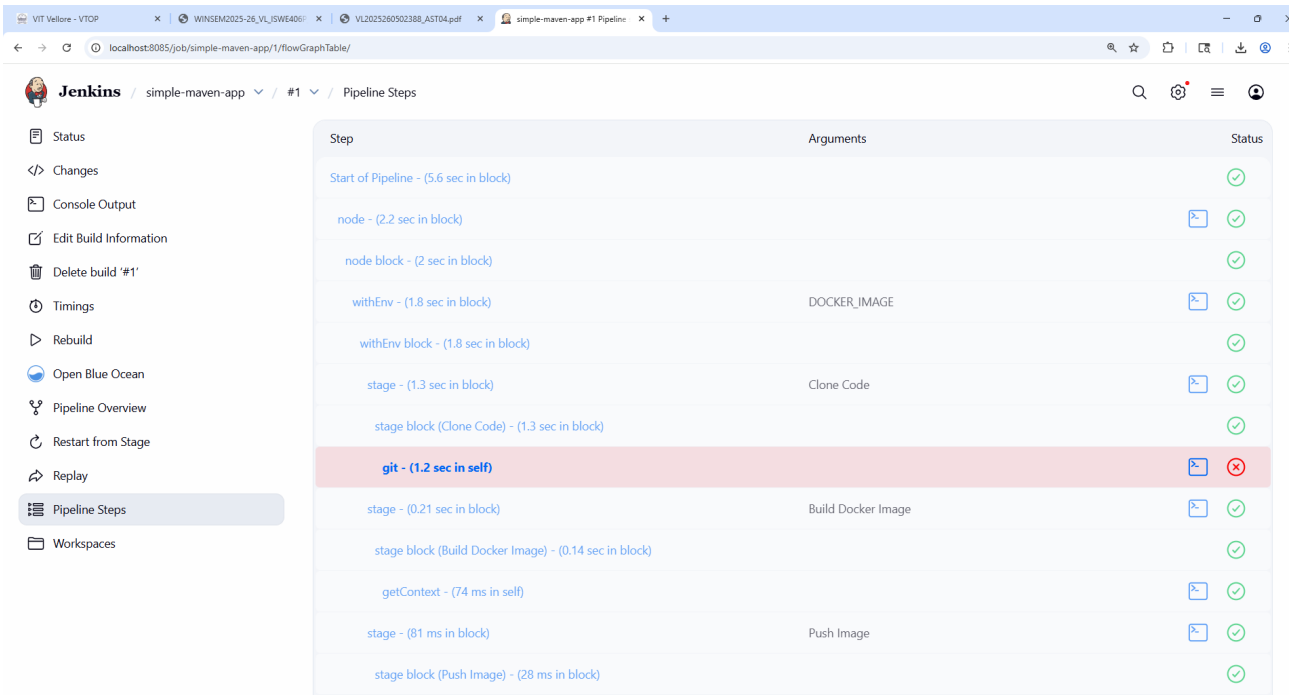
connect GITHUB to JENKINS via PAT token



credential added for DOCKER & KUBERNETES in JENKINS



pipeline connection



ALL SET-Ups Done now Building the desired projects

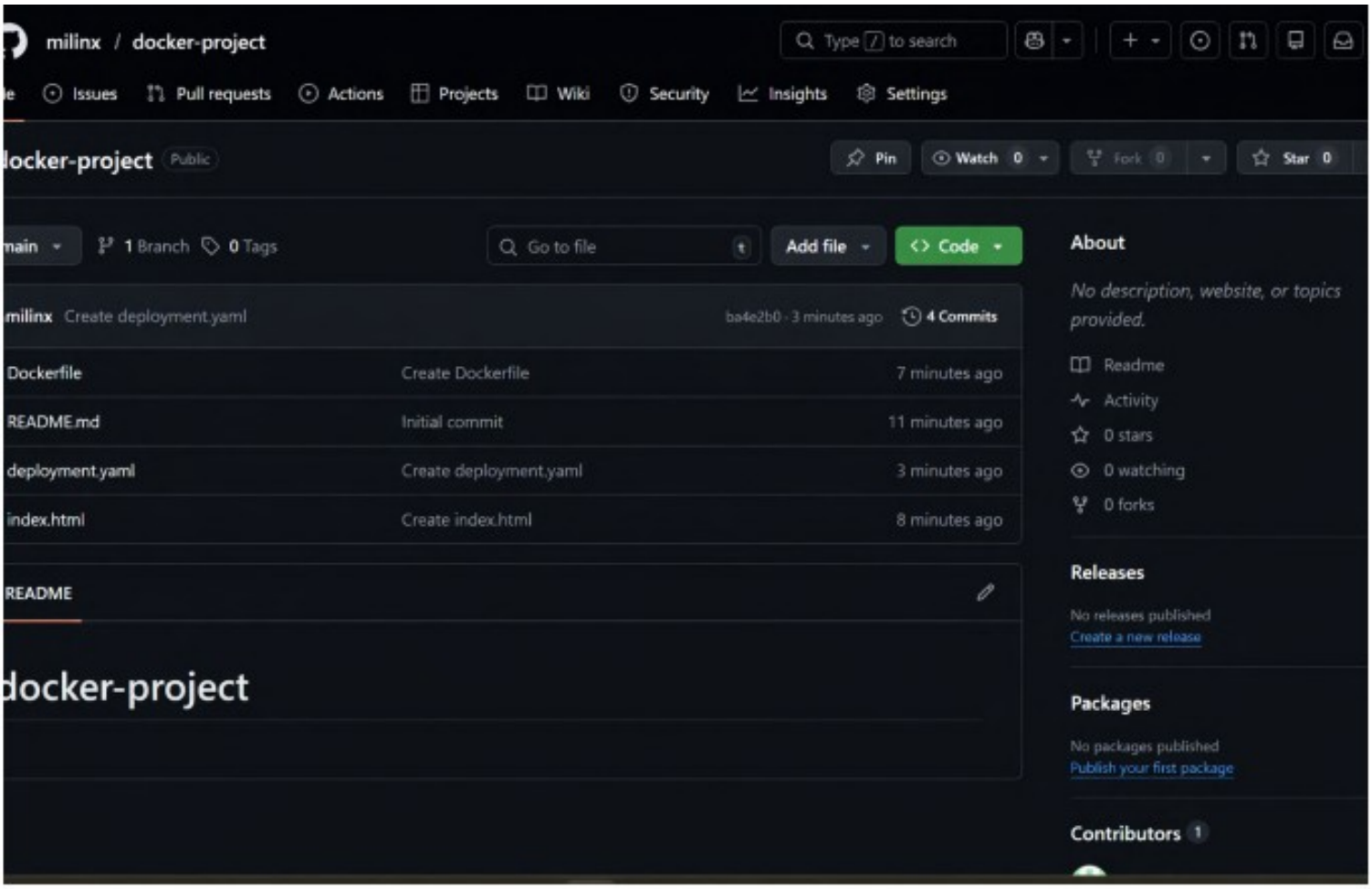
1. Implement CI and CD using GIT+GITHUB+DOCKER + KUBERNETECS using JENKINS for any application.

Create a repository in GIT and create the following files:

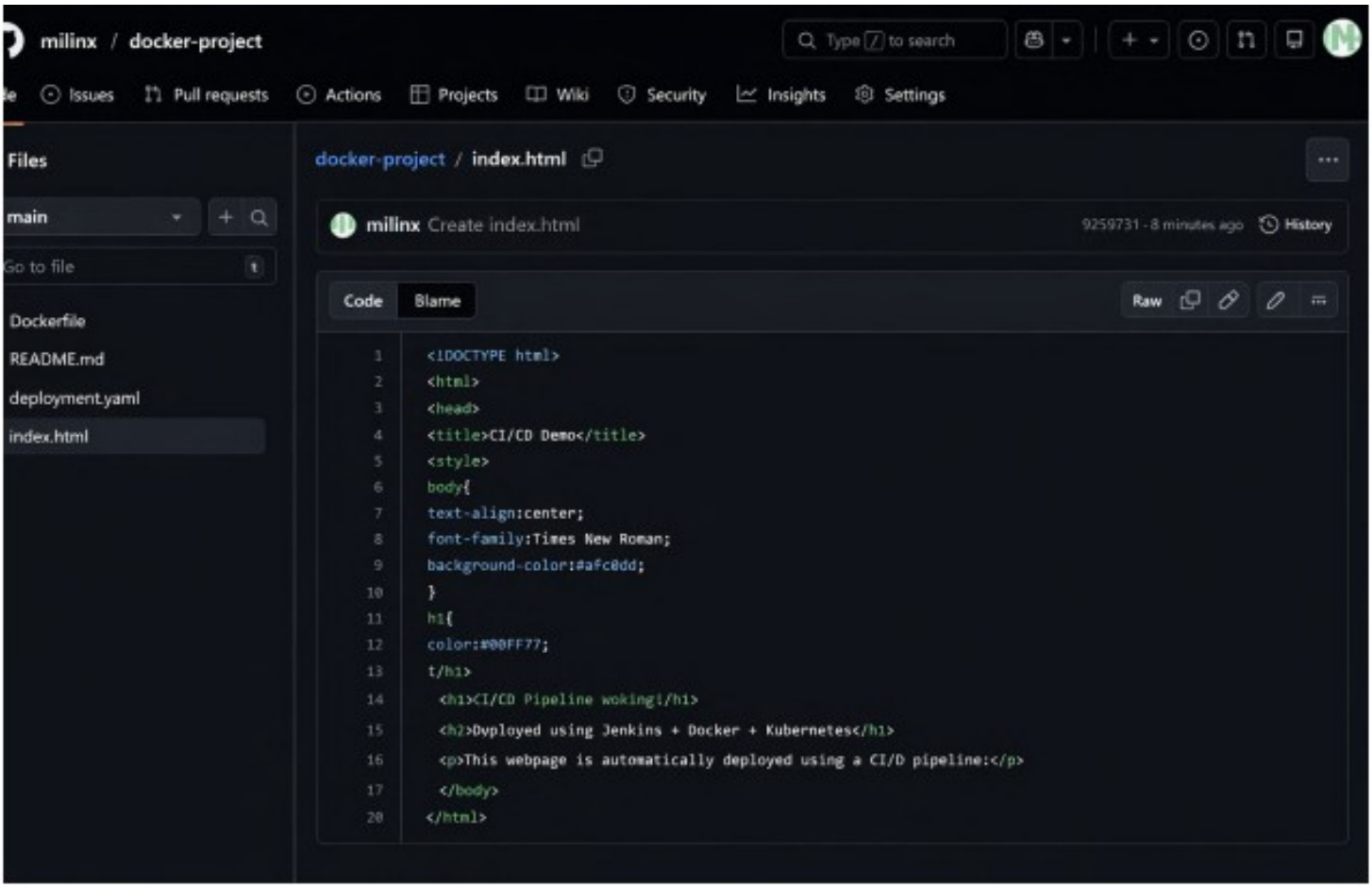
•index.html

•Dockerfile

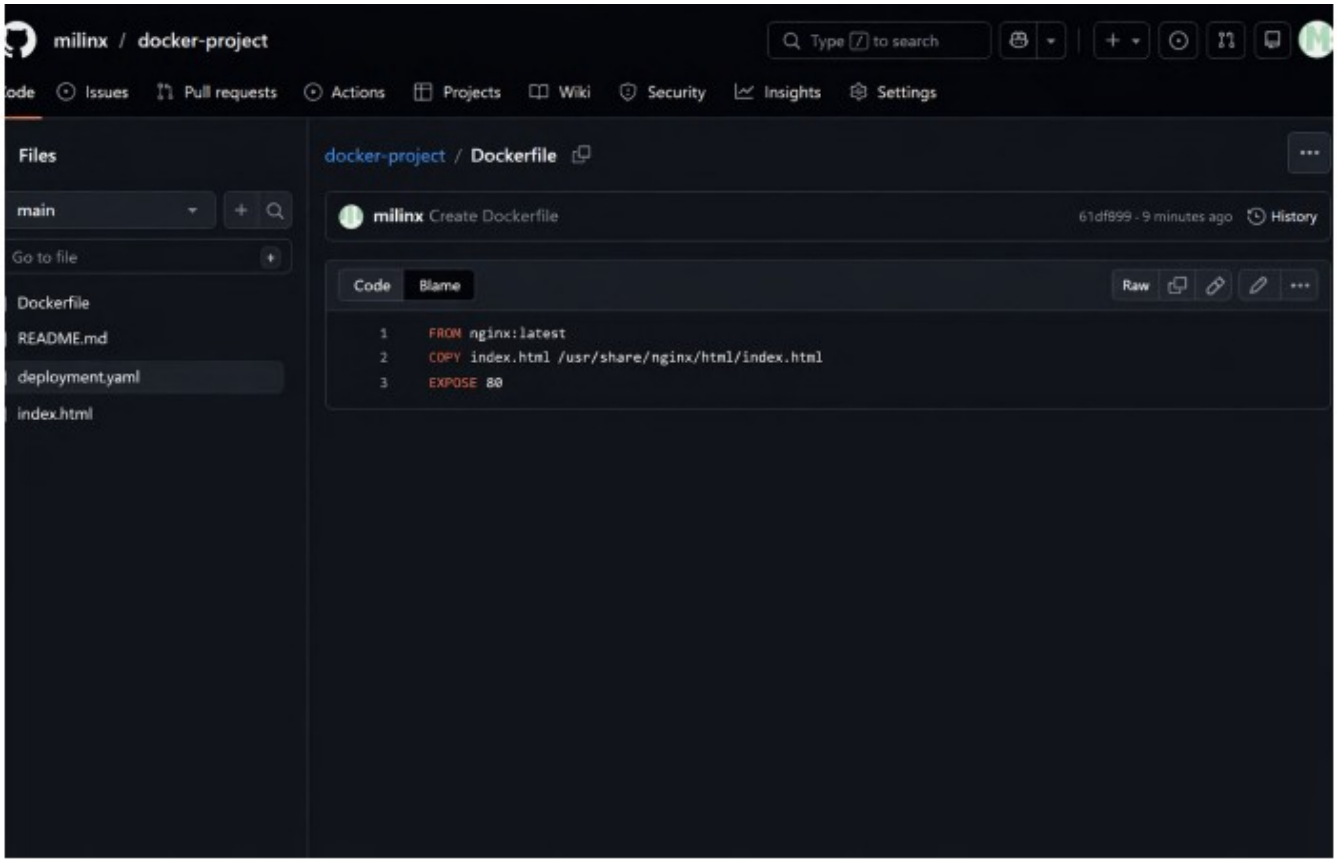
•Deployment.yaml



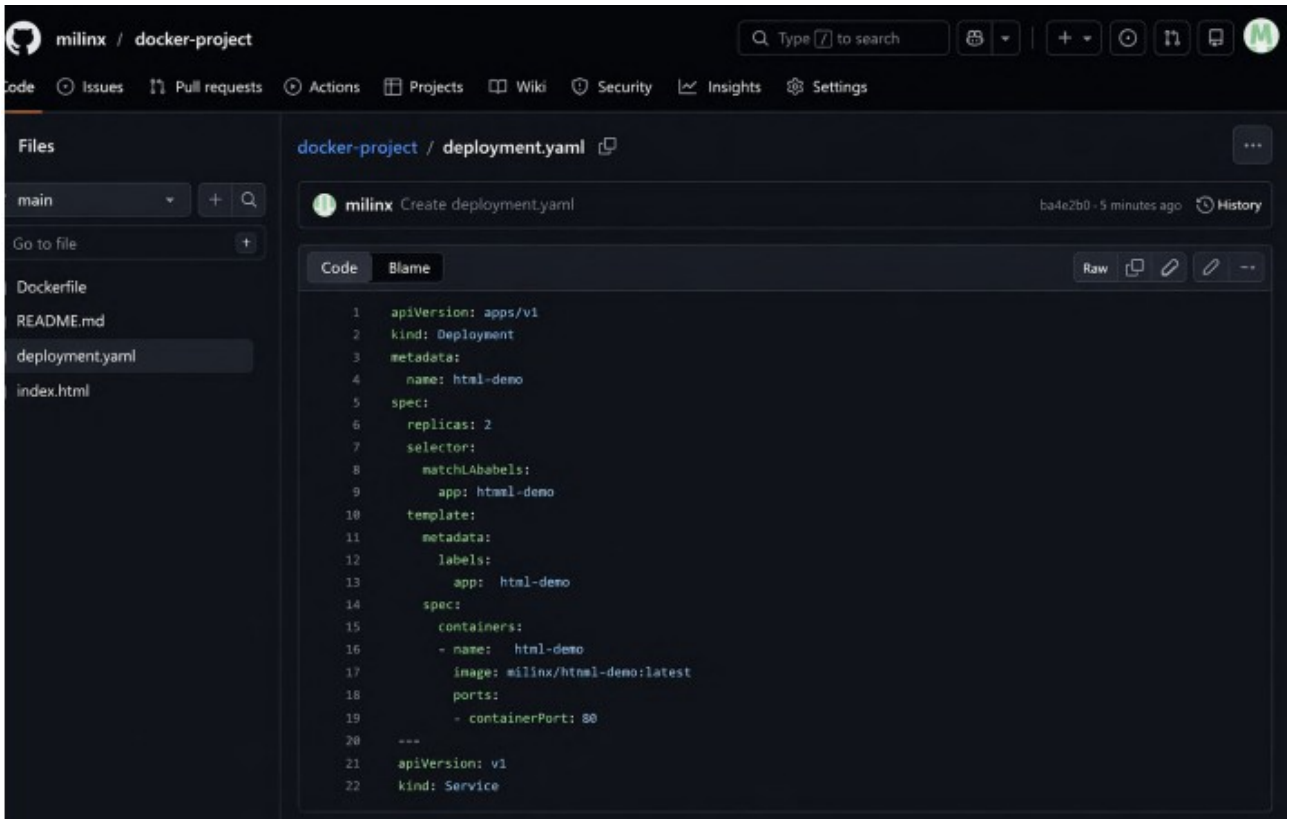
INDEX, HTML



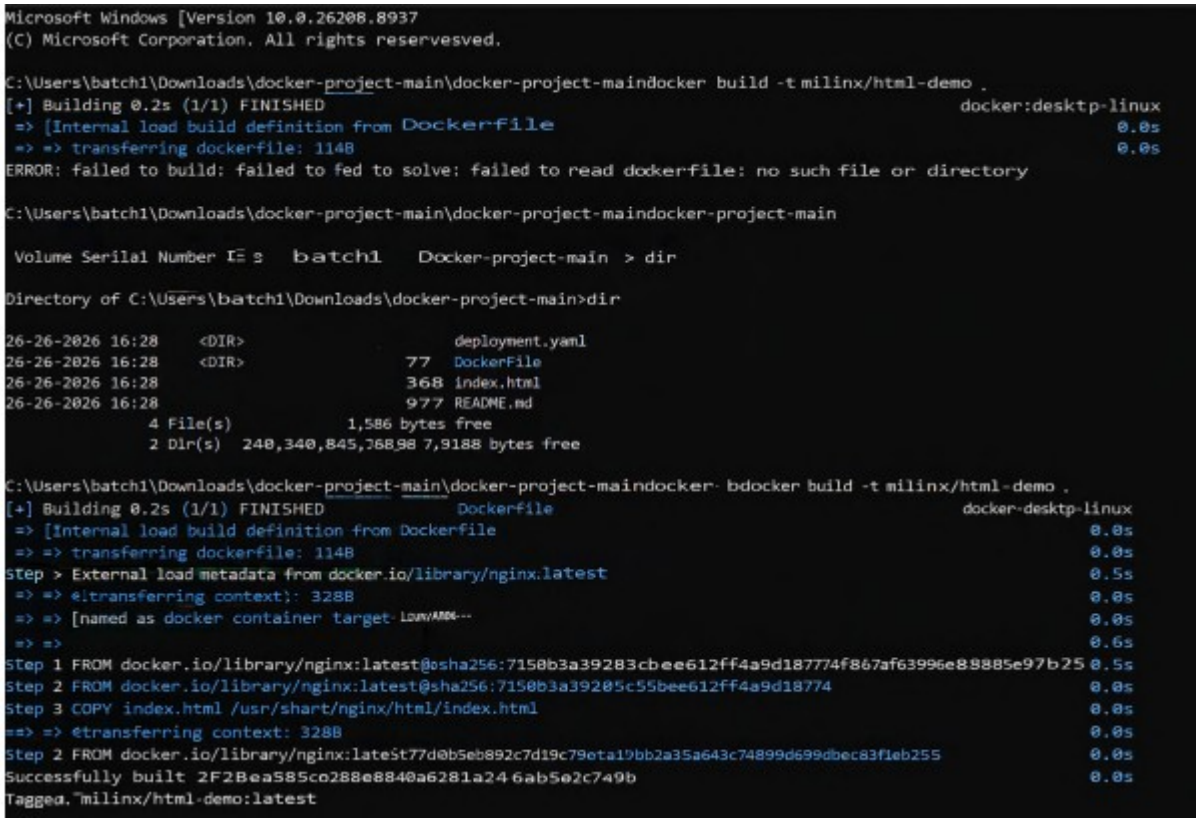
DOCKERFILE:



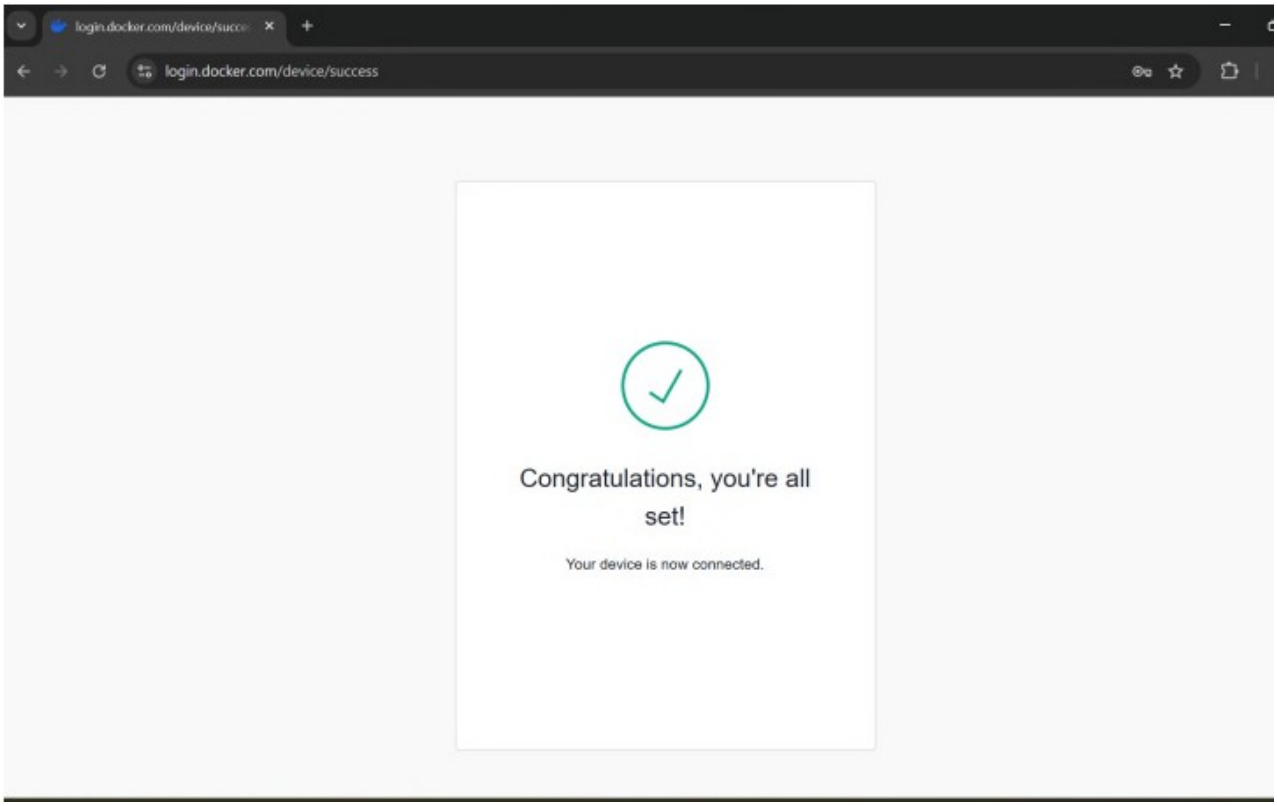
DEPLOYMENT.YAML:



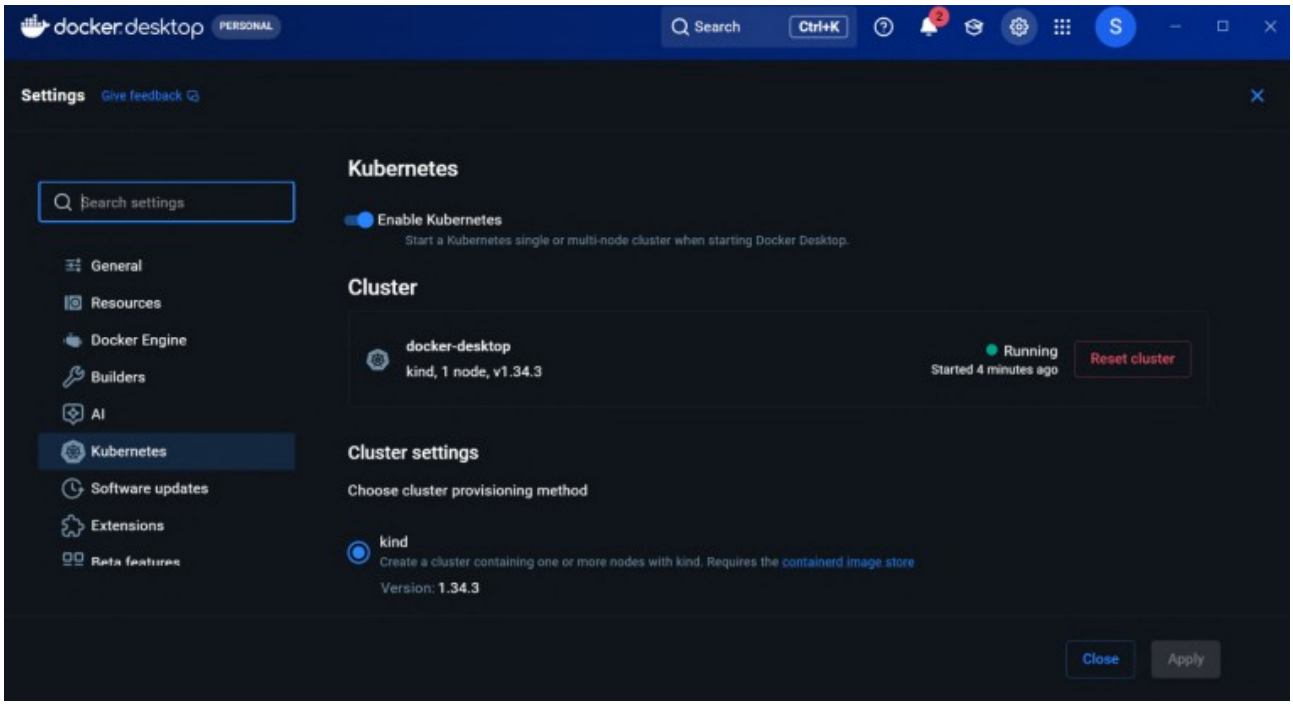
2. Enable Kubernetes

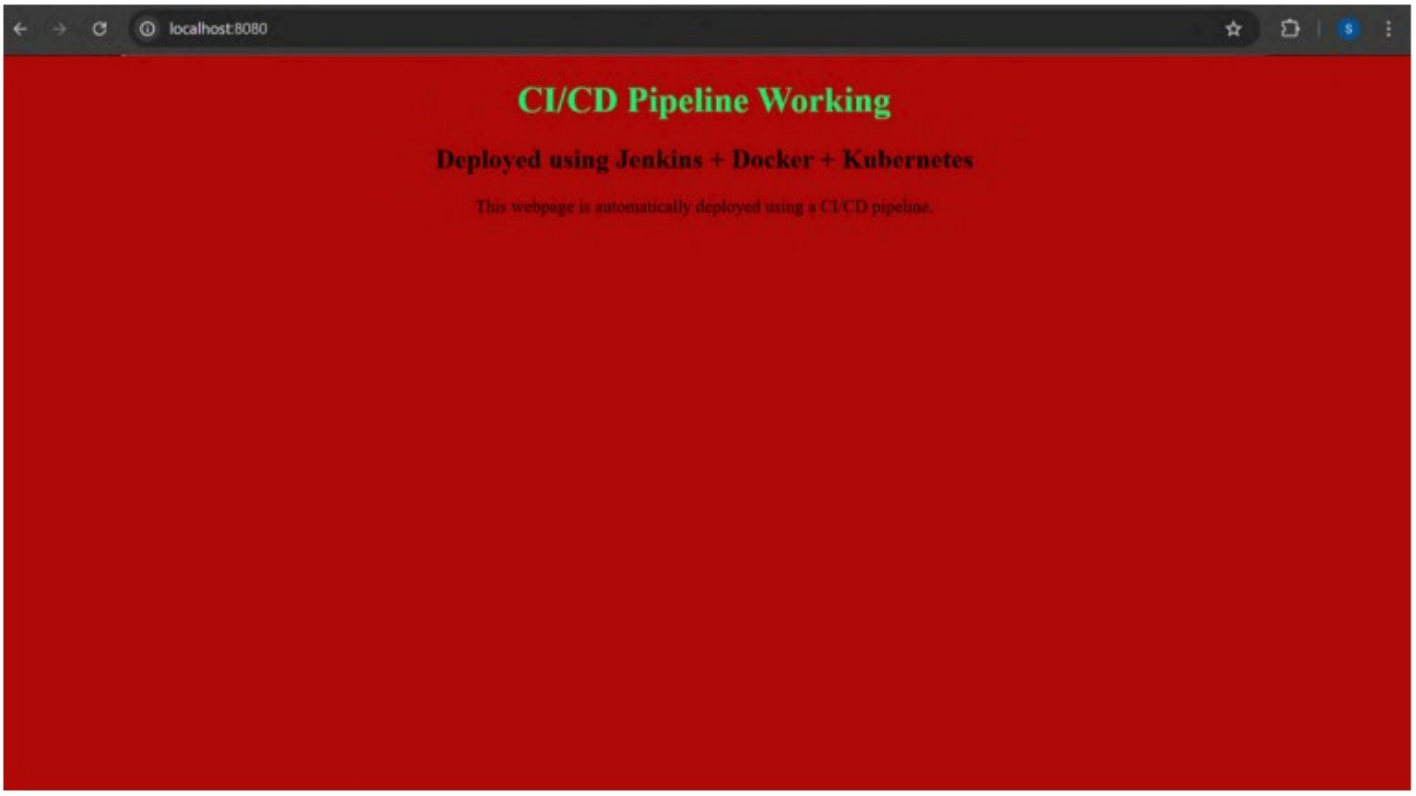


```
0.3s => COPY index.html /usr/share/nginx/html/lndex.html                                   0.3s
0.1s => exporting layers                                                                    0.1s
0.1s => exporting image                                                                     0.1s
0.1s => exporting image sha256:0a411d1d4115abbe5067c74196e75ef47481855e8819258b44e42466     0.1s
0.1s => exporting manifest sha256:44f509931329a731t0b41e2e58868112662943af63b1126f19765     0.1s
0.1s => exporting attestation manifest sha256:a26k7710065c5dec255eb237846b8923c219d84-…4101a9765   0.0s
0.1s => pushing milinx/html-demo:latest sha266:2f2bea585c6286cdb840a281a246                 0.0s
0.1s => naming to docker.io/milinx/html-demo:latest                                         0.1s

C:\Users\batchi\Downloads\docker-project-main\docker-project-main>docker login

USING WEB-BASED LOGIN

Tip: Info => To sign inwth credentials on the command Line, use 'docker login -u <username>'

Enter your one-time device confirmation code is: GSZV-PCZZ
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Opening https://login.docker.com/activate in the browser…

Login Succeeded

C:\Users\batchi\Downloads\docker-project-main\docker-project-main>docker push milinx/html-demo

The push refers to repository [docker.io/milinx/html-demo]
 0176687420519: Pushed    size: 856
 cc57c6e0fac1e: Mounted from library /nginx
 63776706c6449: Mounted from library/nginx
 e39861b07cc2:  Mounted from library/nginx
 cf6e1e2458fb2: Mounted from library/nginx
 13be08f38d08d: Mounted from library/nginx
 2a16db66ce5c0: Mounted from library/nginx
 a9302568ed82i: Mounted from library/nginx
 latest: digest: sha256:2f2bea583c628ebb840a281324246abb502c7498b452fc3Cb008ac4 size: 856

C:\Users\batchi\Downloads\docker-project-main\docker-project-main>
```

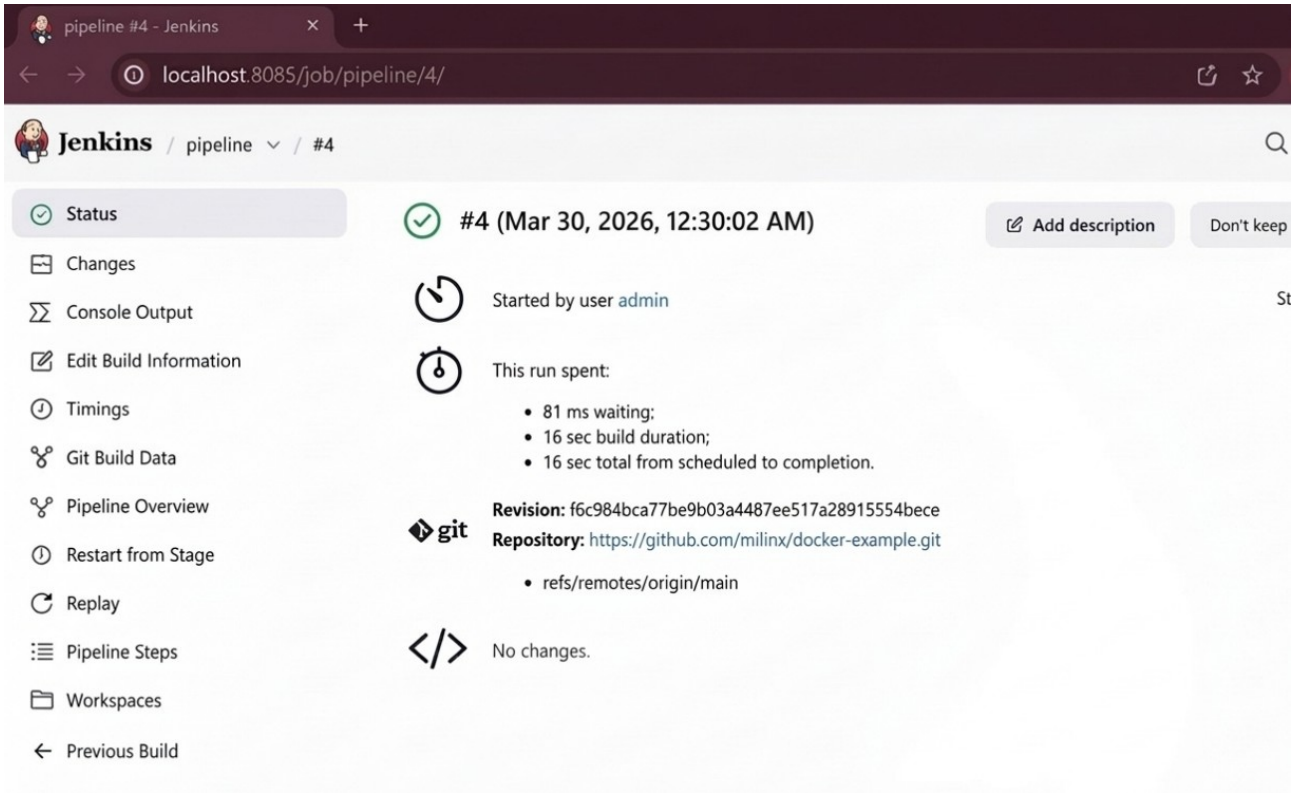


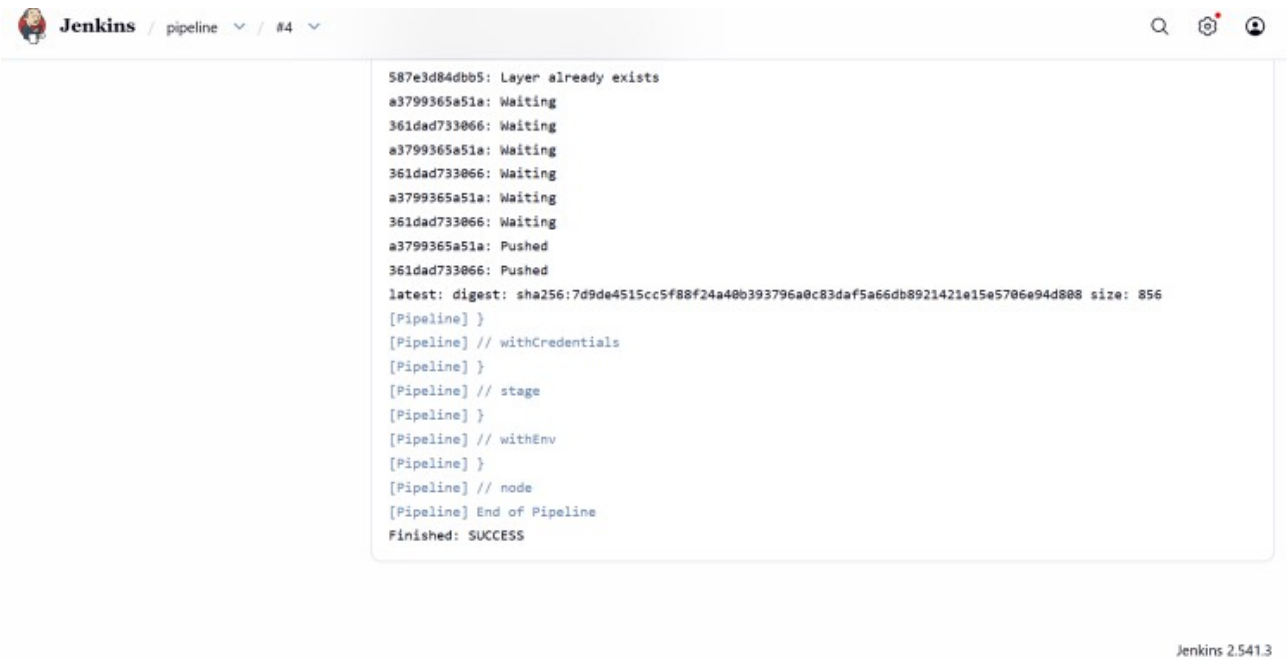
```
C:\Windows\System32\cmd.exe - kubectl port-forward service/html-demo-service 8080:80
C:\Users\batch1\Downloads\docker-project-main\docker-project-main
Volume in drive C is OS
Volume Serial Number is C3A1-534E

Directory of C:\Users\batch1\Downloads\docker-main\docker-project-main

26-3-2026  10:28    <DIR>          .
26-3-2026  10:28    <DIR>          ..
26-3-2026  10:28               521 deployment.yaml
26-3-2026  10:28                77 Dockerfile
26-3-2026  10:28               368 index.html
26-3-2026  10:28                16 README.md
             4 File(s)            5C7 bytes
             2 Dir(s)  67,106,619,392 bytes free

C:\Users\batch1\Downloads\docker-project-main\kubectl port-forward serkubectl apply -f deployment.yaml
html-demo-epΓ11-demo created
html-demo-service created

C:\Users\batch1\Downloads\docker-project-main\kubectl get pods
html-demo-service created
ME     TYPE     STATUS       STATUS   RESTARTS   AGE
demo-service   Running     Running 12s        12s

C:\Users\batch1\Downloads\docker-project-main\kubectl get svc
         TYPE          CLUSTER-IP   EXTERNAL-IP    PORT(S)
html-demo-service      10.96.167.23  <none>        80:30010/TCP
html-demo-Icbs  648a58  14/TCP        <none>        443/TCP

C:\Users\batch1\Downloads\docker-project-main\kubectl port-forward service/html-demo-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Handling connection for 8080
Handling connection for 8080

C:\Users\batch1\Downloads\docker-project-main\kubectl_
```





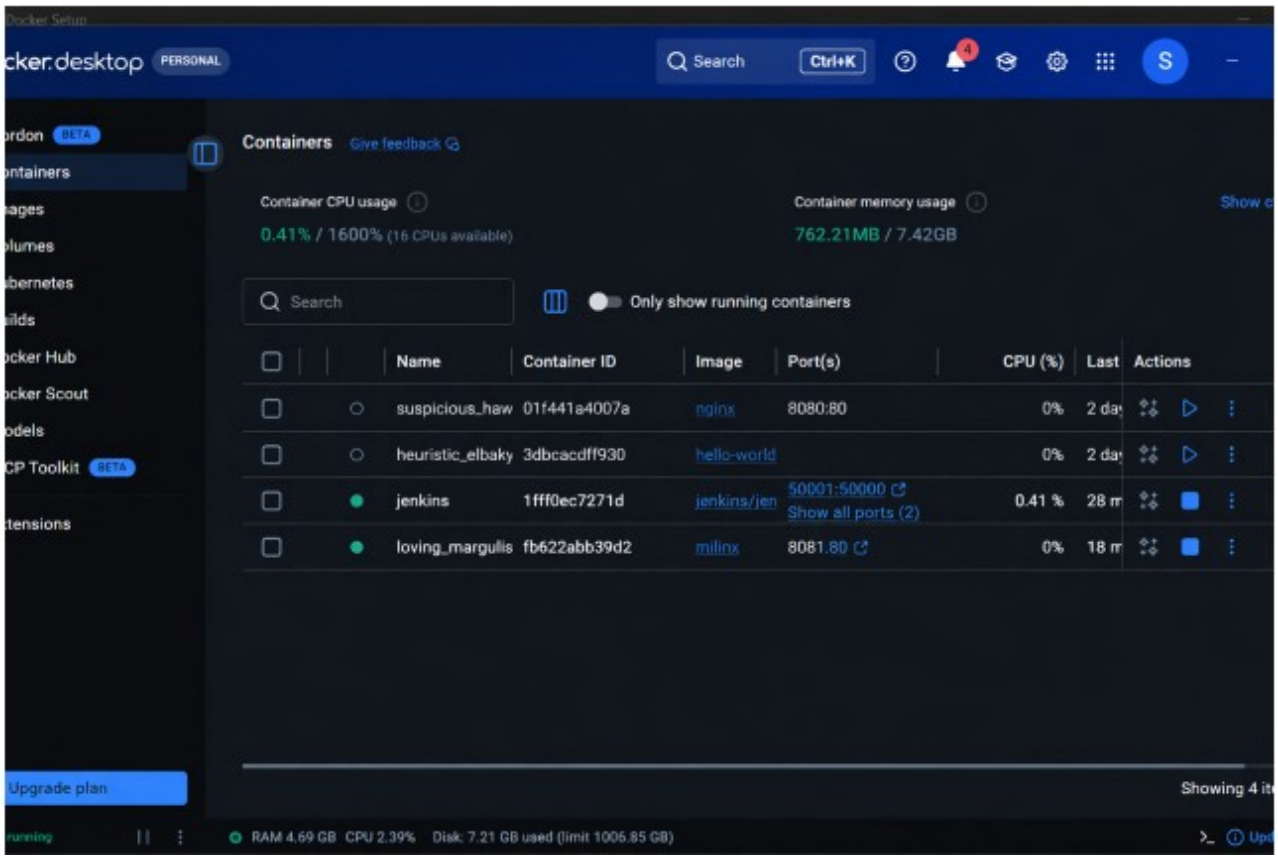
3. Open Jenkins



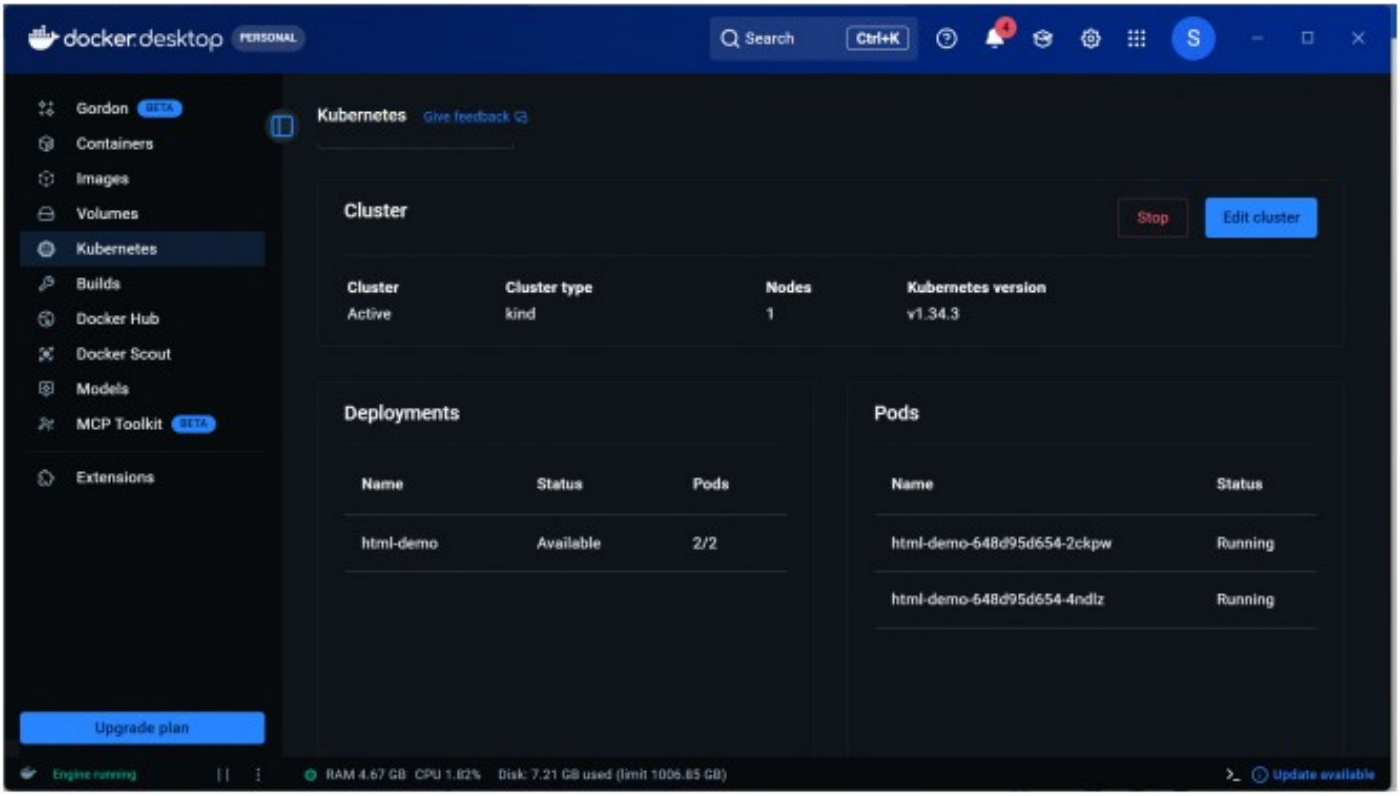


REFLECTION IN DOCKER DESKTOP:

CONTAINERS:

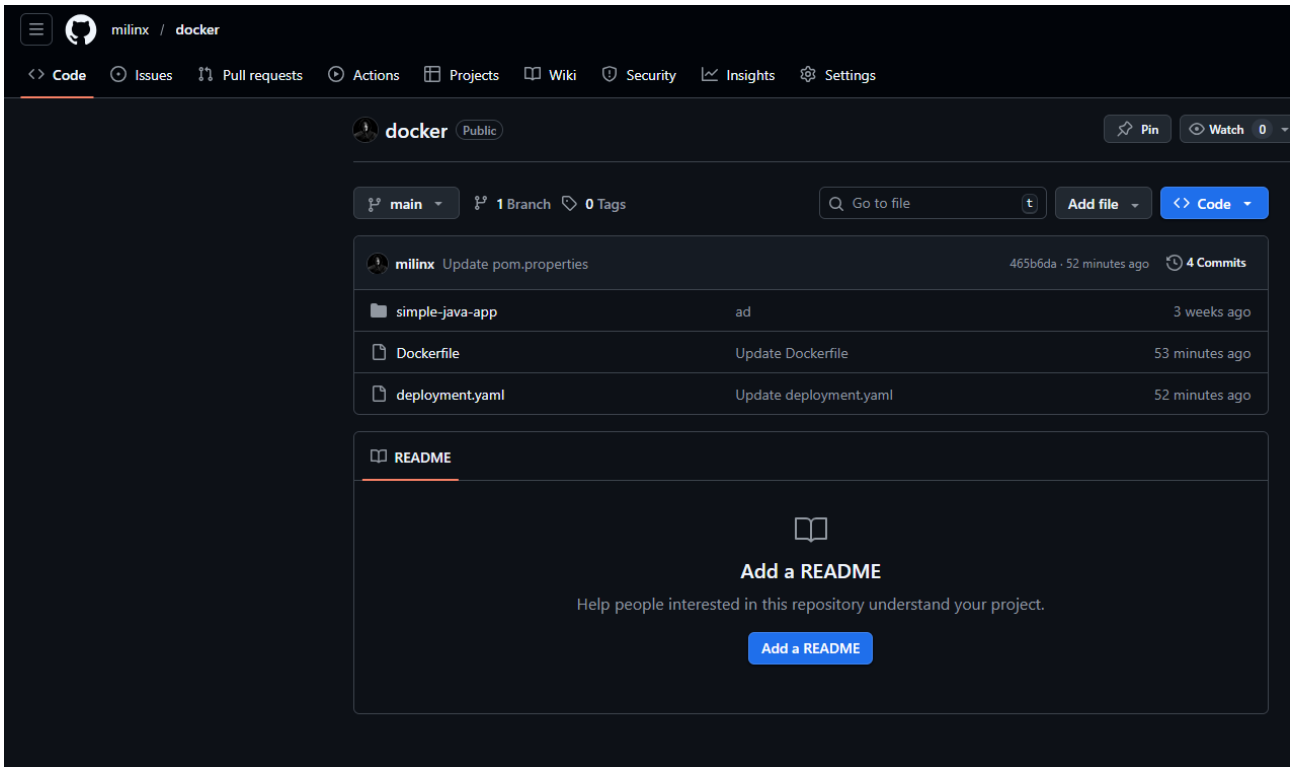


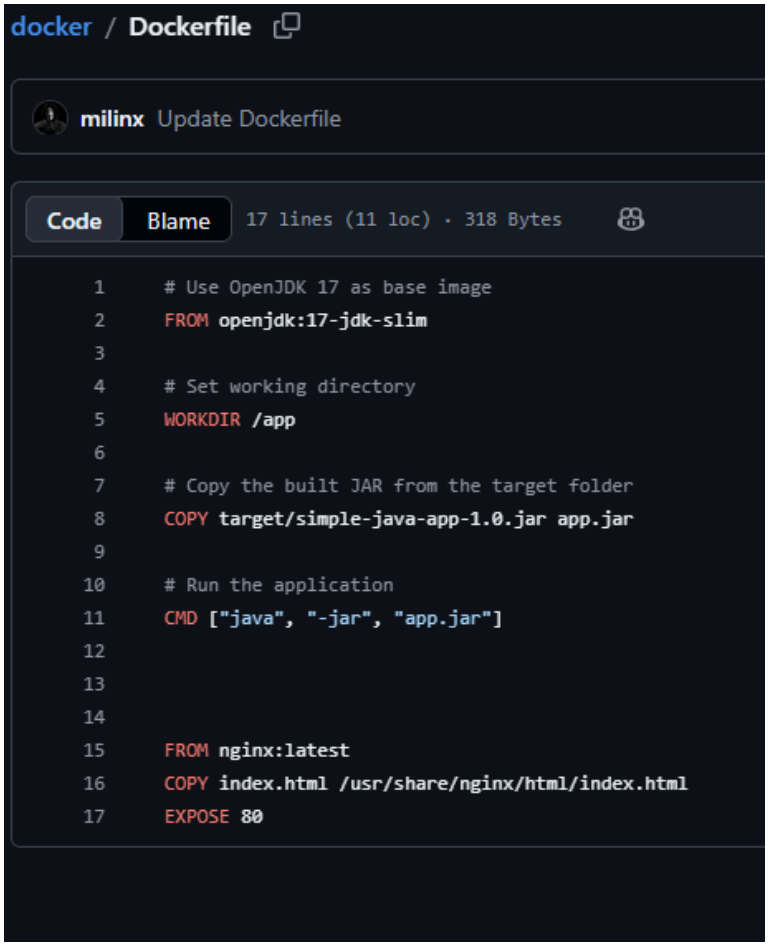
KUBERNETES:



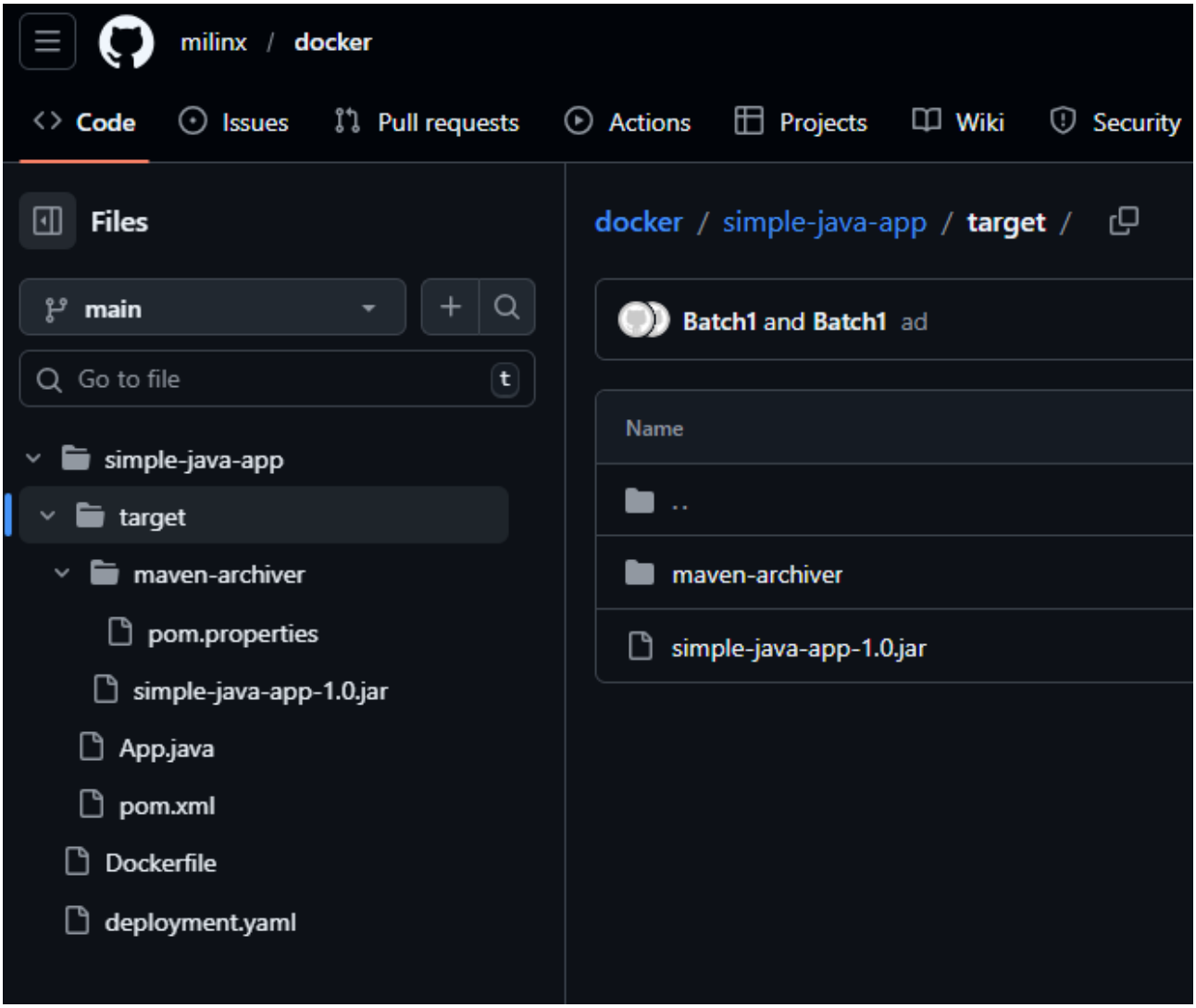
THE HTML FILE OPENS IN *: http://localhost:8085*

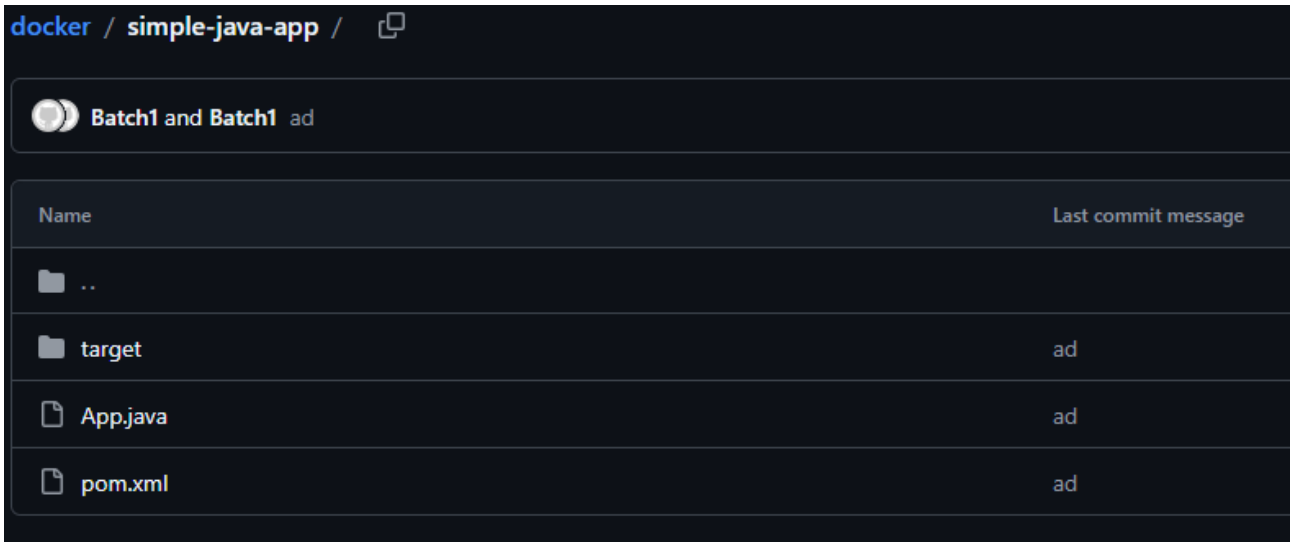
2. Implement CI and CD using GIT+GITHUB+MAVEN+ DOCKER + KUBERNETECS using JENKINS for JAVA application.





Maven :


milinx / docker
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Files
main
Go to file
t
simple-java-app
target
maven-archiver
pom.properties
simple-java-app-1.0.jar
App.java
pom.xml
Dockerfile
deployment.yaml
docker / simple-java-app / target /
Batch1 and Batch1 ad
Name
..
maven-archiver
simple-java-app-1.0.jar


docker / simple-java-app /
Batch1 and Batch1 ad
Name
Last commit message
..
target
ad
App.java
ad
pom.xml
ad

milinx / docker

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

Files

main

Go to file

- simple-java-app
  - target
    - maven-archiver
      - pom.properties
    - simple-java-app-1.0.jar
  - App.java
  - pom.xml
  - Dockerfile
  - deployment.yaml

docker / simple-java-app / pom.xml

Batch1 and Batch1 ad

Code | Blame | 32 lines (29 loc) · 1.1 KB

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.demo</groupId>
    <artifactId>simple-java-app</artifactId>
    <version>1.0</version>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.3.0</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>App</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>
        </plugins>
    </build>
</project>
```

milinx / docker

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

Files

main

Go to file

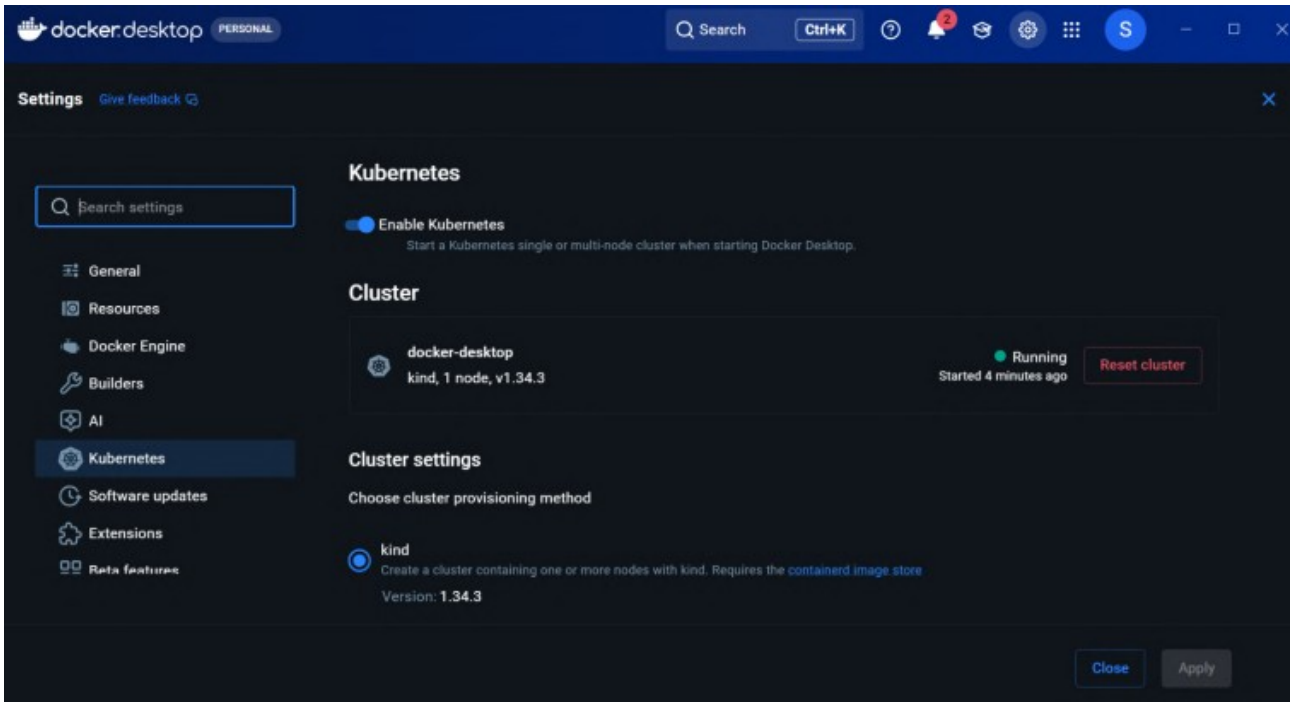
- simple-java-app
  - target
    - maven-archiver
      - pom.properties
    - simple-java-app-1.0.jar
  - App.java
  - pom.xml
  - Dockerfile
  - deployment.yaml

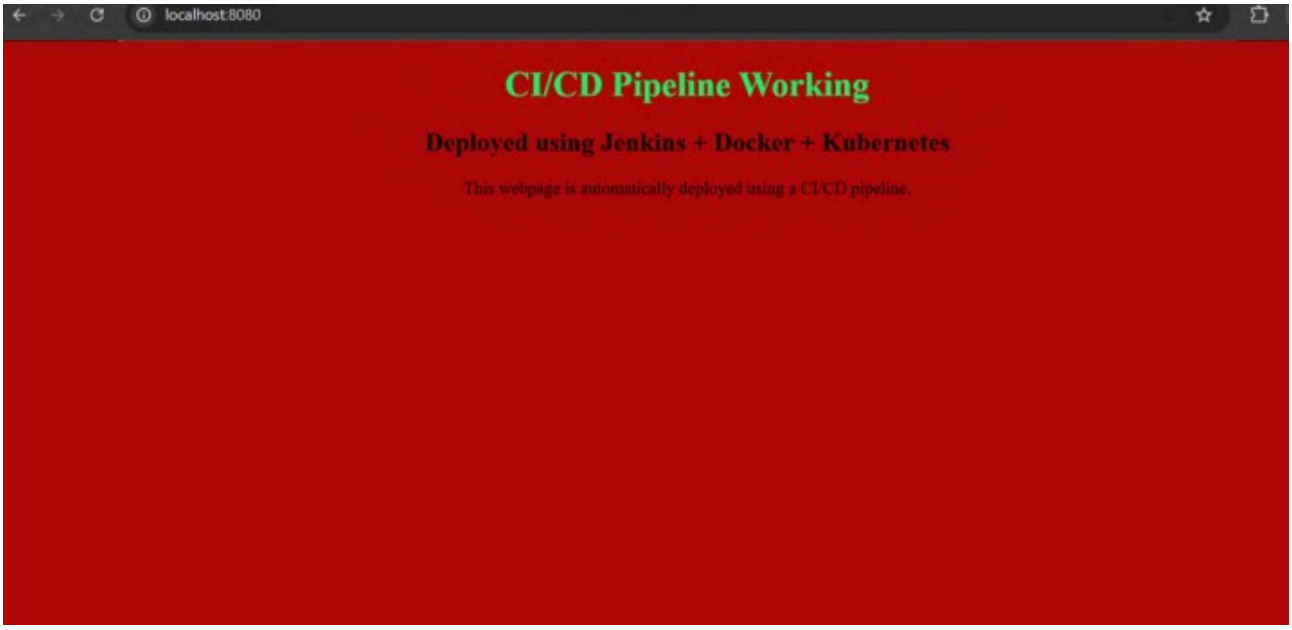
docker / simple-java-app / App.java

Batch1 and Batch1 ad

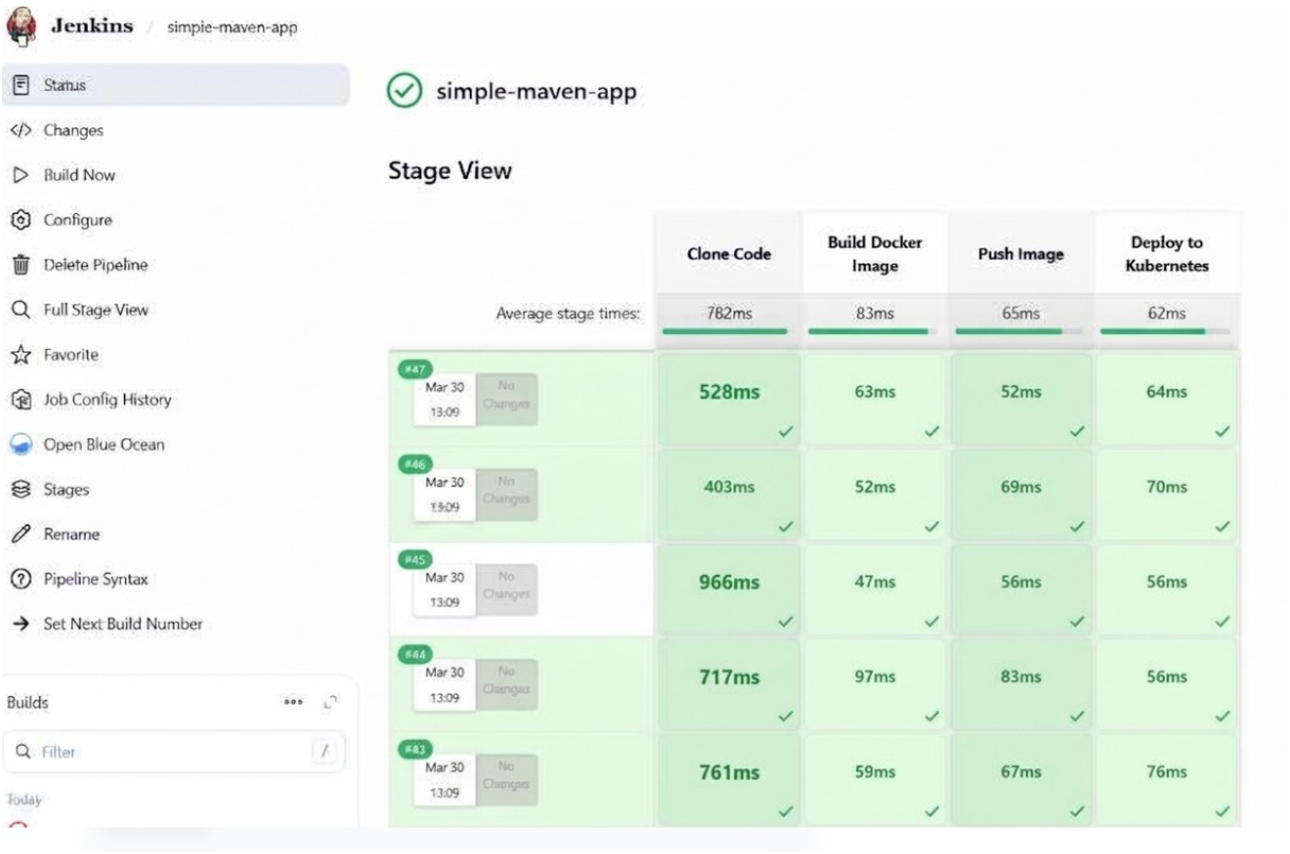
Code | Blame | 33 lines (32 loc) · 969 Bytes

```
public class App {
    public static void main(String[] args) throws Exception {
    System.out.println("Java CI/CD Application Started...");
    while (true) {
    Thread.sleep(5000);
    System.out.println("Hello from Java CI/CD Pipeline!");
    }
    }
   }
   Create Maven File pom.xml
   <project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.demo</groupId>
    <artifactId>simple-java-app</artifactId>
    <version>1.0</version>
    <build>
    <plugins>
    <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-compiler-plugin</artifactId>
    <version>3.10.1</version>
    <configuration>
    <source>17</source>
    <target>17</target>
    </configuration>
    </plugin>
    </plugins>
    </build>
   </project>
```






```
C:\Users\batch1\Downloads\docker-project-main\docker-project-main
Volume in drive C is OS
Volume Serial Number is C3A1-534E

Directory of C:\Users\batch1\Downloads\docker-main\docker-project-main

26-3-2026  10:28    <DIR>          .
26-3-2026  10:28    <DIR>          ..
26-3-2026  10:28               521 deployment.yaml
26-3-2026  10:28                77 Dockerfile
26-3-2026  10:28               368 index.html
26-3-2026  10:28                16 README.md
             4 File(s)           5C7 bytes
             2 Dir(s)  67,106,619,392 bytes free

C:\Users\batch1\Downloads\docker-project-main\kubectl port-forward serkubectl apply -f deployment.yaml
html-demo-epΓ11-demo created
html-demo-service created

C:\Users\batch1\Downloads\docker-project-main\kubectl get pods
html-demo-service created
ME    TYPE     STATUS       STATUS  RESTARTS   AGE
demo-service   Running     Running 12s        12s

C:\Users\batch1\Downloads\docker-project-main\kubectl get svc
         TYPE          CLUSTER-IP   EXTERNAL-IP    PORT(S)
html-demo-service      10.96.167.23  <none>        80:30010/TCP
```

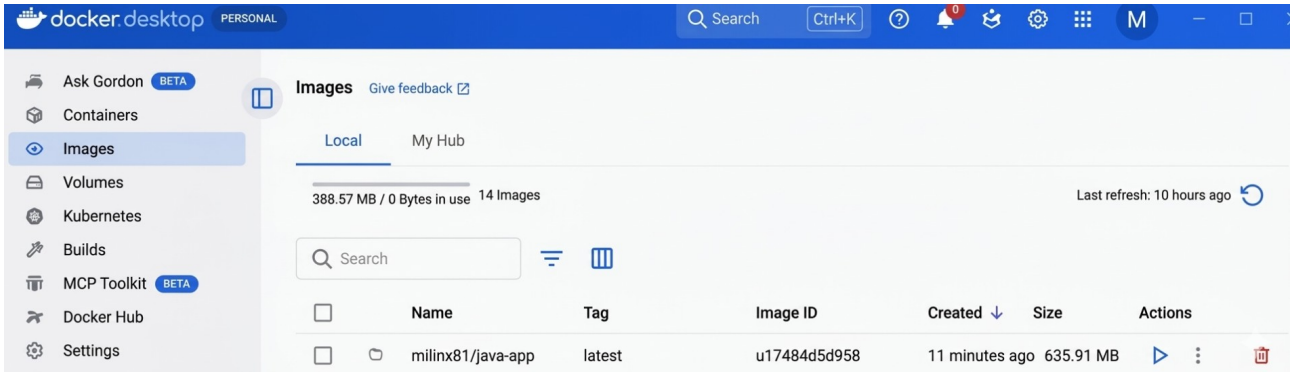



```
deployment.apps/java-app created
service/java-service created
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

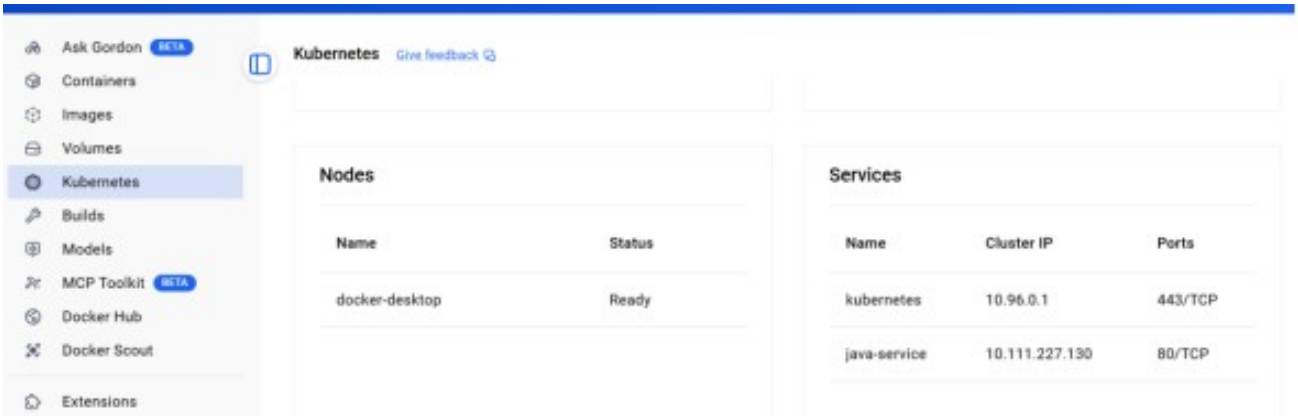
Permalinks

- Last build (#7), 1 min 9 sec ago
- Last failed build (#7), 1 min 9 sec ago
- Last unsuccessful build (#7), 1 min 9 sec ago
- Last completed build (#7), 1 min 9 sec ago

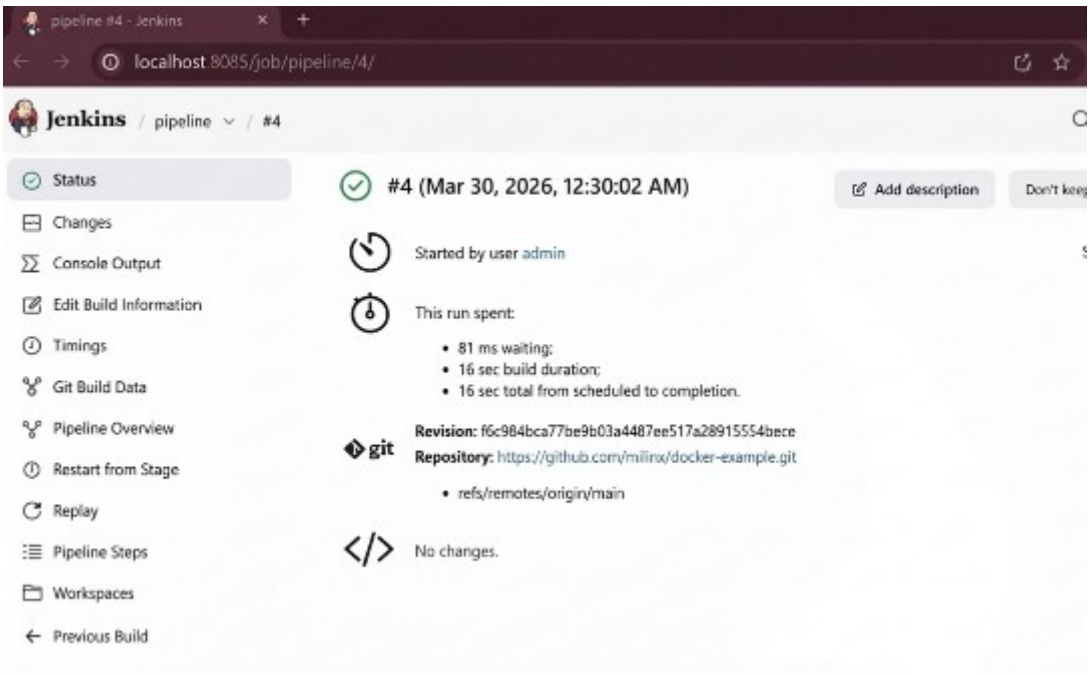
Docker Image



Kubernetes




```
PS C:\Users\batch1\simple-java-app> kubectl get svc
NAME             TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
java-service     NodePort    10.111.227.130  <none>        80:30008/TCP   5m2
kubernetes        ClusterIP  10.96.0.1       <none>        443/TCP        169
PS C:\Users\batch1\simple-java-app>
```


pipeline #4 - Jenkins
localhost:8085/job/pipeline/4/
Jenkins / pipeline / #4
Status
Changes
Console Output
Edit Build Information
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#4 (Mar 30, 2026, 12:30:02 AM)
Add description
Started by user admin
This run spent:
81 ms waiting;
16 sec build duration;
16 sec total from scheduled to completion.
git
Revision: f6c984bca77be9b03a4487ee517a28915554bece
Repository: https://github.com/milinx/docker-example.git
refs/remotes/origin/main
No changes.